Asthra

Asthma Digital Twin Platform

System Architecture

Data Sources
(Synthetic / Open Source)

1. Electronic Health Records (Synthea - FHIR R4)

FHIR

- Patient demographics
- Diagnosis history
- Medications
- Hospitalization records
- Allergies & comorbidities

2. Physiological Time-Series (Wearables / IoT Sensors) - Simulated Data)

- Peak Expiratory Flow (PEF)
- SpO_2
- Heart Rate
- Respiratory Rate
- Symptoms (self-reported)

3. Environmental Data (Air Quality & Weather) - Simulated Data)

- PM2.5 / PM10
- Pollen levels
- Temperature
- Humidity
- Other atmsspheric factors

Data Ingestion & Processing

- Data Extraction & Parsing
- Data Cleaning & Normalization
- Time Alignment & Synchronization
- Data Quality Checks
  - Missing values
  - Out of range values
  - Stale data detection
  - Sensor consistency

Patient Digital Twin Engine

Patient Registry (FHIR-based)

- Unique patient ID
- Clinical profile
- Baseline information

PatientTwin Instance (Per Patient)

- Personalized baseline
- Rolling medians & MAD
- State isolation
- Real-time updates
- Provisional vs. established baseline

Feature Engineering & Model Pipeline

Feature Engineering

- Time-aware features
- Rolling statistics
- Environmental lag features
- Symptom patterns
- Historical EHR features

ML Models (scikit-learn)

- Logistic Regression (EHR baseline)
- HistGradientBoostingClassifier (Dynamic & Fused Model)
- Rule-based baseline
- Prediction explanations (feature importance)

Deterioration Risk Prediction (6-Hour Forecast)

- P(PEF < 80% of personal baseline)
- Uncertainty estimation
- Safe abstention on low-quality data

Backend API Server
(FastAPI)

FastAPI

- Patient management APIs
- Data ingestion APIs
- Prediction APIs
- Environment analysis APIs
- What-if simulation APIs
- Historical data APIs
- Model evaluation APIs

Frontend
(React Dashboard)

React

- Patient Selection & Profile View
- Real-time Monitoring (Charts & Trends)
- Deterioration Risk & Predictions
- Environmental Trigger Analysis
- What-If Scenario Simulation
- EHR & History View

Clinician / Researcher

Data Storage Layer (SQLite + SQLAlchemy)

Structured storage for all data and model outputs

- Patients (EHR Data)
- Observations (Time-Series)
- Environmental Data
- Predictions & Alerts
- Simulation Results

Outputs & Insights

- Personalized Respiratory Monitoring
- Early Deterioration Risk Alerts
- Environmental Trigger Analysis
- What-If Scenario Simulation Results
- Prediction Explanations